DSC5001: Statistical Machine Learning I

Tutorial 8: Discriminant Analysis, Logistic Regression, and Evaluation

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Task 1: LDA with Correlated Sensors

A factory uses two standardized measurements to detect equipment faults: x_1 is vibration and x_2 is temperature deviation. Class 0 means normal and class 1 means faulty.
Assume a Gaussian model with shared covariance:

$$\boldsymbol{\mu}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\mu}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \pi_0 = \frac{3}{4},\ \pi_1 = \frac{1}{4}.$$

For a new machine, $\boldsymbol{x}_* = \left(\frac{5}{2}, 1\right)^\top$. Use the LDA scores

$$\delta_k(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k + \log \pi_k.$$

(a) Compute $\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.

(b) Find the score difference $D(\boldsymbol{x}) = \delta_1(\boldsymbol{x}) - \delta_0(\boldsymbol{x})$, then the posterior probability $P(Y = 1 \mid \boldsymbol{x}_*)$.

(c) Derive the equal-cost decision boundary and classify $\boldsymbol{x}_*$. Find the boundary again if both class priors are $1/2$.

(d) Explain why x_2 cancels from this particular LDA rule and how the unequal priors affect the decision.

Solution: Task 1(a) – Covariance Inverse

The covariance matrix has determinant

$$|\boldsymbol{\Sigma}| = 2(2) - 1(1) = 3.$$

Its inverse is

$$\boxed{\boldsymbol{\Sigma}^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.}$$

The difference between the class means is

$$\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Therefore,

$$\boxed{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.}$$

Solution: Task 1(b) – Posterior Probability

Subtract the two LDA scores. Since $\boldsymbol{\mu}_0 = \mathbf{0}$,

$$\begin{aligned} D(\boldsymbol{x}) &= \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - \tfrac{1}{2} \boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 + \log \frac{\pi_1}{\pi_0} \\ &= x_1 - 1 - \log 3. \end{aligned}$$

For $\boldsymbol{x}_* = (2.5, 1)^\top$,

$$\boxed{D(\boldsymbol{x}_*) = 1.5 - \log 3 \approx 0.4014.}$$

Because D is the log posterior odds,

$$P(Y = 1 \mid \boldsymbol{x}_*) = \frac{1}{1 + e^{-D(\boldsymbol{x}_*)}}.$$

Thus $\boxed{P(Y = 1 \mid \boldsymbol{x}_*) \approx 0.5990}$.

Solution: Task 1(c) – Decision Boundary

For equal classification costs, predict class 1 when

$$P(Y = 1 \mid \boldsymbol{x}) > \frac{1}{2} \quad \Longleftrightarrow \quad D(\boldsymbol{x}) > 0.$$

Here $D(\boldsymbol{x}) = x_1 - 1 - \log 3$. The boundary is

$$\boxed{x_1 = 1 + \log 3 \approx 2.099.}$$

Since the new machine has $x_1 = 2.5$, its predicted class is

$$\boxed{\widehat{y}_* = 1 \quad \text{(faulty)}.}$$

With equal priors, $\log(\pi_1/\pi_0) = 0$, so the boundary moves to

$$\boxed{x_1 = 1.}$$

Solution: Task 1(d) – Interpretation

The discriminating coefficient vector is

$$\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Hence this particular LDA score difference is $x_1 - 1 - \log 3$: x_2 has no additional effect after accounting for the shared covariance.
This cancellation is **specific to the supplied means and covariance**; it is not true for every LDA problem.

Main idea

With only 25% faulty machines, the classifier needs stronger evidence to predict a fault. The unequal priors move the boundary from 1 to approximately 2.099.

Task 2: QDA with Unequal Class Spreads

A quality-control sensor gives a real-valued score X for products from two sources, $Y = 0$ and $Y = 1$. Their score distributions are

$$X \mid Y = 0 \sim \mathcal{N}(0, 1), \qquad X \mid Y = 1 \sim \mathcal{N}(2, 4).$$

Source 0 supplies one-third of products and source 1 supplies two-thirds: $\pi_0 = 1/3$, $\pi_1 = 2/3$.
For QDA, the log posterior odds can be written as

$$D(x) = \log \frac{\pi_1}{\pi_0} - \frac{1}{2} \log \frac{\sigma_1^2}{\sigma_0^2} - \frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_0)^2}{2\sigma_0^2}.$$

(a) Substitute the parameters and simplify $D(x)$ to a quadratic expression.

(b) Find every equal-cost decision boundary and the regions predicted as class 1.

(c) Compute $P(Y = 1 \mid X = x)$ at $x = 0$ and $x = 1$.

(d) Explain why an extremely negative sensor score can be classified as source 1, even though that source has mean 2. Can one-feature LDA produce the same two-boundary rule?

Solution: Task 2(a) – QDA Score Difference

Here $\mu_0 = 0$, $\sigma_0^2 = 1$, $\mu_1 = 2$, $\sigma_1^2 = 4$, and $\pi_1/\pi_0 = 2$.

$$\begin{aligned} D(x) &= \log 2 - \tfrac{1}{2}\log 4 - \frac{(x-2)^2}{8} + \frac{x^2}{2} \\ &= \frac{4x^2 - (x-2)^2}{8}. \end{aligned}$$

The log prior ratio and the log standard-deviation ratio cancel.
Expanding gives

$$\boxed{D(x) = \frac{3x^2 + 4x - 4}{8} = \frac{(x+2)(3x-2)}{8}.}$$

Solution: Task 2(b) – Two Decision Boundaries

An equal-cost decision boundary satisfies $D(x) = 0$:

$$(x + 2)(3x - 2) = 0.$$

There are two boundaries:

$$\boxed{x = -2 \qquad \text{and} \qquad x = \frac{2}{3}.}$$

Predict class 1 when $D(x) > 0$, or equivalently

$$(x + 2)(3x - 2) > 0.$$

Therefore,

$$\boxed{\widehat{y} = 1 \text{ for } x < -2 \text{ or } x > \frac{2}{3}.}$$

Predict class 0 for $-2 < x < 2/3$. At the boundaries, the posterior probabilities are tied.

Solution: Task 2(c) – Posterior Probabilities

Convert log posterior odds into probability using

$$P(Y = 1 \mid X = x) = \frac{1}{1 + e^{-D(x)}}.$$

At $x = 0$, $D(0) = -1/2$, so

$$\boxed{P(Y = 1 \mid X = 0) = \frac{1}{1 + e^{1/2}} \approx 0.378.}$$

At $x = 1$, $D(1) = 3/8$, so

$$\boxed{P(Y = 1 \mid X = 1) = \frac{1}{1 + e^{-3/8}} \approx 0.593.}$$

Thus $x = 0$ favors source 0, while $x = 1$ favors source 1.

Solution: Task 2(d) – The Wider Class

Class 1 has standard deviation 2, compared with standard deviation 1 for class 0. Its density has **heavier tails**.
For example, at $x = -3$,

$$D(-3) = \frac{(-3+2)(3(-3)-2)}{8} = \frac{11}{8} > 0.$$

Hence QDA predicts class 1, despite its larger distance from that class's mean.

Main idea

The greater spread of source 1 can outweigh its larger distance from a very unusual score. A one-feature LDA comparison has a linear score difference and cannot generate these two distinct boundaries.

Task 3: Logistic Loss and a Gradient Step

A subscription company models customer churn $Y \in \{0, 1\}$, with $Y = 1$ for churn. The binary features are $x_1 = 1$ for a recent complaint and $x_2 = 1$ for receiving a retention discount.
Three training cases have features and labels:

Case	x_1	x_2	y
A	1	0	1
B	0	1	0
C	1	1	1

The current model is $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, $p = \sigma(z) = 1/(1 + e^{-z})$, with $\boldsymbol{\beta} = (0, \log 3, -\log 3)^\top$.

(a) Calculate the predicted churn probability for each case.

(b) Calculate the *summed* binary cross-entropy $\mathcal{L} = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$.

(c) Find $\nabla_{\boldsymbol{\beta}} \mathcal{L} = \sum_i (p_i - y_i)(1, x_{i1}, x_{i2})^\top$.

(d) Take one gradient-descent step with learning rate $\rho = 0.2$. Find the new probability for case C and interpret its change.

Solution: Task 3(a) – Logistic Probabilities

The score is $z_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$, with $\boldsymbol{\beta} = (0, \log 3, -\log 3)^\top$.

Case	Score z_i	Probability p_i
A	$\log 3$	$3/4 = 0.75$
B	$-\log 3$	$1/4 = 0.25$
C	0	$1/2 = 0.50$

Therefore,

$$\boxed{\boldsymbol{p} = (0.75, 0.25, 0.50)^\top.}$$

Solution: Task 3(b) – Cross-Entropy

The observed labels are $\boldsymbol{y} = (1, 0, 1)^\top$.
Each case contributes the probability assigned to its actual label:

Case	A	B	C
Assigned probability	0.75	$1 - 0.25 = 0.75$	0.50

The likelihood is

$$L = (0.75)(0.75)(0.50) = \frac{9}{32}.$$

The summed negative log-likelihood is

$$\boxed{\mathcal{L} = -\log \frac{9}{32} = \log \frac{32}{9} \approx 1.269.}$$

Solution: Task 3(c) – Batch Gradient

The probability errors are $\boldsymbol{p} - \boldsymbol{y} = (-1/4, 1/4, -1/2)^\top$.
Including the intercept, the design matrix is

$$\boldsymbol{X} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Hence

$$\boxed{\nabla_{\boldsymbol{\beta}} \mathcal{L} = \boldsymbol{X}^\top (\boldsymbol{p} - \boldsymbol{y}) = \begin{bmatrix} -1/2 \\ -3/4 \\ -1/4 \end{bmatrix}.}$$

The gradient is for the *sum* of losses, not their average.

Solution: Task 3(d) – Updated Prediction

One gradient step with $\rho = 0.2$ gives

$$\begin{aligned}\boldsymbol{\beta}^{\text{new}} &= \boldsymbol{\beta} - 0.2\nabla\mathcal{L} \\ &= \boxed{\begin{bmatrix} 0.10 \\ \log 3 + 0.15 \\ -\log 3 + 0.05 \end{bmatrix}}.\end{aligned}$$

For case C, $(x_1, x_2) = (1, 1)$, so the updated score is

$$z_C^{\text{new}} = 0.10 + (\log 3 + 0.15) + (-\log 3 + 0.05) = 0.30.$$

Hence

$$\boxed{p_C^{\text{new}} = \sigma(0.30) \approx 0.574.}$$

The model now assigns a higher probability to case C's observed positive label.

Task 4: Threshold Costs and ROC AUC

An already-trained fraud detector produces the following *validation-set* probabilities. Fraud is class 1. No model refitting is allowed during this task.

Case	1	2	3	4	5	6	7	8
Actual y	1	0	1	0	0	1	0	0
Score p	.82	.74	.61	.56	.45	.38	.30	.12

Predict fraud whenever $p \geq t$. A false alarm costs 2 units and a missed fraud costs 9 units; correct decisions cost zero.

(a) At thresholds $t = 0.50$ and $t = 0.35$, find TP, FP, FN, and TN.

(b) Compute precision, recall, and accuracy at both thresholds. Comment on the accuracies.

(c) Calculate the total decision cost for each threshold. Which is preferable *among these two* on this validation sample?

(d) Calculate ROC AUC by comparing all positive–negative score pairs. Does changing t change AUC?

Solution: Task 4(a) – Confusion Counts

There are three actual fraud cases: $1, 3, 6$; the other five are negative.
At $t = 0.50$, predicted positives are cases $1, 2, 3, 4$. At $t = 0.35$, predicted positives are cases $1, 2, 3, 4, 5, 6$.

Threshold	TP	FP	FN	TN
0.50	2	2	1	3
0.35	3	3	0	2

Solution: Task 4(b) – Performance Measures

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad \text{Accuracy} = \frac{TP + TN}{8}.$$

Threshold	Precision	Recall	Accuracy
0.50	$2/4 = 0.50$	$2/3 \approx 0.667$	$5/8 = 0.625$
0.35	$3/6 = 0.50$	$3/3 = 1$	$5/8 = 0.625$

Both thresholds have the same accuracy, despite different errors.

Solution: Task 4(c) – Decision Costs

A false positive costs 2 and a false negative costs 9. Thus

$$\text{Total cost} = 2(FP) + 9(FN).$$

At threshold 0.50,

$$\boxed{2(2) + 9(1) = 13.}$$

At threshold 0.35,

$$\boxed{2(3) + 9(0) = 6.}$$

Main idea

Choose $\boxed{t = 0.35}$ among the two candidates for these validation data and costs. Their identical accuracies conceal a large cost difference; final performance must be assessed on unseen test data.

Solution: Task 4(d) – ROC AUC

The positive scores are $0.82, 0.61, 0.38$; the negative scores are $0.74, 0.56, 0.45, 0.30, 0.12$.
Count how many negative scores each positive score exceeds:

Positive score	0.82	0.61	0.38
Negative scores beaten	5	4	2

There are $3 \times 5 = 15$ positive–negative pairs, with $5 + 4 + 2 = 11$ correctly ranked pairs and no ties.

$$\boxed{\text{ROC AUC} = \frac{11}{15} \approx 0.733.}$$

AUC depends on the *ordering of the scores*, not the threshold. Changing t leaves it unchanged.

Tutorial 8: classify, calculate, and evaluate

Questions and discussion